An Exploration Dynamical Systems Chaos Argyris C02

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textbf2 Preliminaries

If therefore, those cultivators of the physical science from whom the intelligent public deduce their conception of the physicist . . . are led in pursuit of the arcanas of science to the study of the singularities and instabilities, rather than the continuities and stabilities of things, the promotion of natural knowledge may tend to remove that prejudice in favour of determinism which seems to arise from assuming that the physical science of the future is a mere magnified image of that of the past.

James Clerk Maxwell, 1873

This book is an attempt to convey concepts of methods evolved in the field of nonlinear dynamics to budding physicists and engineers and to illustrate them using simple examples. The basis for these new ideas on dynamics is the topological or geometrical view of temporal processes which leads to a representation in phase space. In the following, we shall restrict ourselves mainly to those non-linear processes of evolution, the dynamics of which can be described by ordinary differential equations or difference equations. We consider systems with the following characteristics:

i they are finite-dimensional, i.e. each state is determined by a point in a finitedimensional phase space;

ii they are differentiable, i.e. the temporal change of state is modelled by differentiable functions;

iii they are deterministic, i.e. the equations contain no stochastic terms. Moreover, the initial state determines the solution uniquely at least locally. This means in particular that two trajectories can never intersect.

The differential-topological method offers the opportunity of using the structure of these differential equations to investigate them geometrically in phase space. What is novel in our present-day understanding is that we do not concentrate on a single trajectory but consider the solutions as a whole and analyse their stability globally. In the case of dissipative systems, we observe the ensemble of all the trajectories and classify the long-term behaviour via the stability behaviour of individual groups. This type of analysis is called qualitative dynamics or, in mathematical terminology, the modern theory of dynamical systems. The logical consequence is the investigation of the stability of states of equilibrium or periodic solutions, for example, and, moreover, the study of bifurcations which occur for critical control parameters where the dynamics abruptly undergoes structural changes. The decisive factor here is the phenomenon of instability. This plays the central role in the theory of dynamical systems. Instabilities are responsible for the creation of new temporal patterns as a result of bifurcations and furthermore for the occurrence of irregular motions. The consequence is that slight changes in the initial conditions or the system parameters can lead to unpredictable magnification effects. This revolutionary and at the same time startling discovery is fundamental and undermines some supposedly irrefutable principles of science (Cvitanović et al., 2008); this is the subject of the following section.

Prior to this, we would like to point out explicitly which systems are excluded in the following considerations as a consequence of the above-mentioned three properties. These are, in particular, partial differential equations, systems with time delay as well as differential equations with an additional stochastic term which occur for example in synergetics, which we address briefly in section 6.8. In addition, it would go far beyond the scope of this book to discuss dynamical systems which are only piece-wise continuous or piece-wise smooth, i.e. systems which contain jump discontinuities in the function or in the derivative. This includes, for example, mechanical systems with dry friction or systems where shocks or pulses occur, systems with electronic switches, biological systems with thresholds or economic systems. In practice, such systems occur often and can cause complex dynamical behaviour as well as intricate bifurcation scenarios. Recently, these systems have become the subject of intensive research, see (Zhusubaliyev and Mosekilde, 2003; di Bernardo et al., 2008).

2.1 Causality – Determinism

Classical physics was based on the assumption that the future is determined by the present and that, in this way, an accurate knowledge of the present can reveal the future. This belief in a fundamental limitless predictability was expressed in perhaps the most elegant and comprehensible way by Laplace in 1814 through his fictional demon:

"Une intelligence qui pour un instant donné, connaîtrait toutes les forces dont la nature est animée, et la situation respective des êtres qui la composent, si d'ailleurs elle était assez vaste pour soumettre ces données à l'analyse, embrasserait dans la même formule, les mouvements des plus grands corps de l'univers et ceux du plus léger atome: rien ne serait incertain pour elle, et l'avenir comme le passé, serait présent à ses yeux." (Laplace, 1812)

Anyone expressing the certitude of natural laws so pointedly, who, indeed, interprets causality so narrowly as to mean determinism, is asking to be contradicted, particularly at a time when the consequences of the quantum theory and the development of atomic physics affect not only philosophy but our thought processes.

This confrontation makes it clear how generation-dependent the principle of causality or the law of cause and effect is and how it changes in the course of developments in thinking and language. We must conclude that the concept of causality and our understanding of it is linked with the concepts and the meanings associated with them at anyone time! Formulations such as causality, causal law, causal explanation or causal principle are always confusing as long as their explanations are not transparent. In this connection, it appears helpful to begin with the historical development of these concepts.

The fact that causality sets up a link between cause and effect is, historically speaking, relatively recent. For Aristotle, the originator of many scientific disciplines, the word causa had a much more general meaning than nowadays. He expressed his thoughts in the two pithy concepts "xxxxx". With direct reference to Aristotle, scholasticism, the philosophy of the Middle Ages, taught that there are are four types of cause: 1. causa materialis (= material cause); 2. causa formalis (= formal cause); 3. causa finalis (= final cause); 4. causa efficiens (= efficient cause). In addition, Aristotle differentiated between internal and external causes, the first two mentioned above being internal, the second two external. The one we roughly associate with the word "cause" nowadays is causa efficiens, the initiating or efficient cause.

With the emergence of the Renaissance and the birth of modern scientific thinking – usually associated with names such as Nicolaus Copernicus, Galileo Galilei and Johannes Kepler – a change in the concept of causa also came about. As man turned away from metaphysics and towards physics, i.e. towards quantification and metrology, the word causa was linked with the material incident which preceded and somehow caused the phenomenon to be explained. At first, it was held that if the cause had been discovered, the natural phenomenon could be explained. In this connection, cause was synonymous with explanation. Gradually, however, the understanding took hold that it was the natural laws, the interconnection of cause and effect, which clearly determine and thus explain natural processes. The culmination of this development was the epoch-making work of Isaac Newton. Immanuel Kant, who, in his "Critique of Pure Reason", equated science as such with the science of Newton, found a formulation for causality which was still valid in the 19th century: "Wenn wir erfahren, daß etwas geschieht, so setzen wir dabei jederzeit voraus, daß etwas vorhergehe, woraus es nach einer Regel folgt."

Einstein praised Newton's outstanding intellectual achievement as "perhaps the greatest progress in thinking that a single individual ever had the privilege to accomplish". His most eminent achievement was his comprehensive mathematical theory of mechanics which remained the basis of scientific thinking until well into the 20th century. It is hardly surprising that his general laws of motion – obeyed by all objects in the solar system from the apple falling from the tree to the planets – or their universal applicability fed the expectation that processes in nature were in principle clearly determined, assuming they are known as a whole or in part after careful study. Newton's universe was a grand mechanical system which functioned in accordance with exact deterministic laws so that the motion of a system in the future could be calculated in advance from the state of this system at any given time. If one is convinced that nature fundamentally behaves in this way, the next logical step is the one Laplace formulated for his fictional supernatural intelligence mentioned above: all the future as well as the past of our universe is calculable from the precise knowledge of the position and velocity of all atoms at any one instant.

This concept of pre-calculability in all eternity is not consistent with 20th century physics and particularly not with quantum mechanics. It is not that physics is in fundamental opposition to man's longing for predictablity but that the two, quantum mechanics and determinism, are incompatible with one another in their philosophical approaches. The atomistic ideas of Democritus and Leucippus handed down from classical antiquity assume that regular, ordered processes on a global level derive their morphology from the irregular, random behaviour on a local level. Such considerations are indeed plausible, as is substantiated by numerous examples from everyday life. A fisherman, for example, fighting against the wind and the waves, only has to ascertain the rhythm of the waves in order to be able to react to them; he does not have to know the motion of each individual drop of water. If we explain the processes we can perceive with our senses by the interaction of many individual processes on the local level, is it not an inevitable consequence to regard the overall regularities of nature as statistical? Statistical laws may provide reliable statements on the whole set of possible events such as, for example, on mean values and possible deviations, yet there will never be a totally reliable prediction for a particular event; this will always remain indeterminate.

In spite of such conceptual difficulties, we are continually setting up statistical laws in day-to-day life as a basis for our practical actions. Engineers, for example – whether they are acting in the design of aircraft, buildings or machines – cannot base their work on precise loads or material data. They have to rely as a matter of course on mean, that is to say statistical characteristic values. Yet nevertheless, when our attention is drawn to such "semi-exact" regularities, we feel uneasy and consider them less trustworthy. We would prefer either precisely definable processes in nature or, on the other hand, chaotic, totally irregular ones. Is there an explanation for such an attitude? One generally uses statistical laws when the physical system in question is only partially known. The simplest example of this is the game of heads or tails. Since no one side of the coin has an advantage over the other, we have to come to terms with the fact that – when playing a large number of games – we can only predict one of the two results with 50possible results has increased to six. The likelihood of predicting a chosen number is thus reduced to 1/6. If the die is thrown often enough, then the number of throws resulting in a 1 is approximately one-sixth of all the throws. This is, of course, only true if we assume a perfect die and an identical throwing technique; only then can all six possible results be considered equally probable.What we assume is that these conditions are satisfied approximately.

In modern times, it was Robert Boyle who took up the idea from classical antiquity by not only describing the material behaviour on the macroscopic level of observation qualitatively as the result of the statistical behaviour of the molecules in a gas but by also quantifying the well-known relationship between pressure and volume. His idea was that the pressure, a quantity measurable on the macroscopic level, was built up by the numerous impacts of molecules on the wall of the vessel; this interpretation was anmiraculous brainwave at the time. In a similar way, the laws of thermodynamics could be conceived when it proved possible to state in a mathematically precise form the fact that atoms move more violently in hot bodies than in cold ones.

While Laplace raised the hopes that one day, the whole world could, in principle, be calculable, in the second half of the last century, the idea spread that Newton's mechanics might be unrestrictedly valid, but that systems within the kinetic gas theory could never be completely determinable due to the immense number of gas molecules. It was mainly Josiah Willard Gibbs and Ludwig Boltzmann who put the incomplete knowledge of these systems into mathematical language by using statistical laws. Gibbs went one step further by introducing temperature as a physical concept for the first time; this only makes sense when the knowledge of the system is incomplete. If indeed the velocity and the position of all the molecules in a gas were known, it would be completely superfluous or meaningless to talk of the temperature of the gas. Hence, the concept of temperature can only be used meaningfully when the system on the microscopic level of observation is incompletely known and one nevertheless does not want to forgo a "qualitative" statement on the macroscopic level. Using such a concept, one does not describe the behaviour of a system by taking an increasing number of degrees of freedom into account – in the ideal case, infinitely many – but by proceeding to new, essential and thus considerably fewer degrees of freedom on a more general level of observation. The motto is not "more precise, more detailed and infinitely many" but "more global, fewer and nevertheless informative". In this highly simplified statement, we share the view of Max Born (1959) who says that absolute accuracy is not a physically meaningful concept and can only be found in the conceptual world of mathematicians. Felix Klein called for the application of "approximation mathematics" side by side with the usual "precision mathematics". Since his suggestion remained without response at the time, the physicists at the turn of the preceding century solved their problems in their own conceptual framework, using methods of probability and statistical laws.What Max Born's statement basically implied was that non-linear laws and deterministic equations may supply unpredictable answers. At the same time, he in effect pointed out that non-linear equations often react with unexpected sensitivity to the slightest changes in the initial conditions and thus suddenly supply unexpectedly differing answers. This was a revolutionary perception, first expressed by Poincar´e at the turn of the preceding century but unnoted for a long time:

"Une cause tr‘es petite, qui nous ´echappe, d´etermine un effet consid´erable que nous ne pouvons pas ne pas voir, et alors nous disons que cet effet est dˆu au hasard. Si nous connaissions exactement les lois de la nature et la situation de l'univers ‘a l'instant initial, nous pourrions pr´edire exactement la situation de ce mˆeme univers ‘a un instant ult´erieur. Mais, lors mˆeme que les lois naturelles n'auraient plus de secret pour nous, nous ne pourrons connaˆıtre la situation initiale qu'approximativement. Si cela nous permet de pr´evoir la situation ult´erieure avec la mˆeme approximation, c'est tout ce qu'il nous faut, nous disons que le ph´enom‘ene a ´et´e pr´evu, qu'il est r´egi par des lois; mais il n'en est pas toujours ainsi, il peut arriver que de petites diff´erences dans les conditions initiales en engendrent de tr‘es grandes dans les ph´enom‘enes finaux; une petite erreur sur les premi‘eres produirait une erreur ´enorme sur les derniers. La pr´ediction devient impossible et nous avons le ph´enom‘ene fortuit." (Poincar´e, 1908)

In the first two decades of the 20th century, the attempt was initally made to explain the motion of the atoms and molecules according to the basic precepts of classical mechanics, in the spirit of Newton. The result, however, was an entanglement of inextricable contradictions. For example, in accordance with classical physics, a charged electron was supposed to orbit around the atomic nucleus and constantly emit radiation until it collapsed into the nucleus due to loss of energy. However, the electron paths devised in the model were unstable. It was obvious to Niels Bohr that, on the basis of Planck's theory, the paths of the electrons are stationary, that the electron, as long as it sticks to its path, does not emit any radiation, but that a change of path is accompanied by a loss of energy. Bohr solved this contradiction not by changing his concept of the model, as could have been expected, but rather by maintaining that classical physics was not applicable to the description of the dynamical behaviour of atomic relations. The fact that loss of energy which takes place in irregular bursts and in stages when an electron changes its path should lead to the assumption that the radiation emittance of atoms is a statistical phenomenon was acceptable. It is also possible to tolerate, at least temporarily, a new type of mechanics, quantum mechanics, albeit at the expense of determinacy, when the stability of the atoms can thus be mathematically ensured. But the bold assertion that it is fundamentally impossible to know all the necessary defining elements in order to achieve a complete determination of the processes – even Albert Einstein could not and would not accept this as true, even for atomic phenomena. For him, the quantum theory was only an instrument, temporarily necessary; it had emerged due to our lack of knowledge of all the canonical variables of the atomic process but could be suspended as soon as all these unknowns had been clarified. His opinion of quantum mechanics culminated in the statement, “God does not play dice”, to which Niels Bohr replied, “It would be presumptuous of us human beings to prescribe to the Almighty how he is to take his decisions.” What is it about the quantum theory that is so challengingly unfamiliar? It is the fact that the concept of the trajectory has been banned from a mechanical theory of the atom shell and replaced by irreducible probabilistic elements. This was imperative in order to ensure discrete energy levels, whose existence had been revealed by spectroscopy, in a mathematical equation for systems such as the atom. It was Werner Heisenberg who took this radical step in 1925 in his decisive work with the title “ ¨Uber quantentheoretische Umdeutungen kinematischer und mechanischer Beziehungen” (Heisenberg, 1925). Here, the canonical variables position and impulse become non-commutative quantities; Max Born and Pascual Jordan recognised their matrix character. Although the fundamental features of this mathematical theory had already been set up, Heisenberg described the problem of atomic physics in a letter to Pauli in 1926 as completely unsolved. What caused him to make this negative statement? What was still missing was the link to the physical experiment (Heisenberg, 1969).

The solution was provided, again by Heisenberg, by the indeterminacy relation in 1927 (Heisenberg, 1927). Position and velocity or momentum, the classical quantities for the determination of a particle trajectory, can be stated individually with arbitrary accuracy; simultaneously, however, this is impossible in the quantum philosophy, at least in the microworld of atomic physics. This is not an assumption in quantum mechanics, but the consequence of the laws of the quantum theory. It was thus clear that Newton’s mechanics, which assumes the exact knowledge of both position and momentum in order to calculate a mechanical course of motion, is not applicable to the atomic world. Although more than seventy years have passed since Bohr, Heisenberg, Born and others came to the conclusion that the quantum theory forces us to formulate the laws on the quantum level as statistical laws and to bid farewell to determinism, it is difficult to incorporate this into our general philosophy. For the atomic field, this call to abandon pure determinism may be valid. But to speak of absolute randomness and to assert that the classical idea of predictability is invalid in principle, although it was so successful in its search for order, regularity and natural laws, merely due to spontaneous nuclear disintegration for which there is neither cause nor explanation – this is something we still cannot and do not want to bring ourselves to accept.

In contrast to this, it should be remembered that classical physics tacitly complements the principle of "identical" causality with the principle of "similar" causality. Laplace's principle, "identical causes have identical effects", was extended by the principle "similar causes have similar effects". The reason is that it is a justifiable expectation to demand exactly identical causes so as to guarantee the reproducibility of physical processes, but that this is not feasible in practice. Any experimenter knows that, although he is at pains to obtain the same results for repeated measurings under the same test conditions, exactly identical repetitions of the test conditions are basically impossible. The accuracy of measurements is limited, even though it may be extremely high with today's technical sophistication. When the errors in the measurement results are of the same order as the inaccuracies of the experimental set-up, they are not exactly the same, but similar. In spite of statistical laws which we take as the basis of our experimenting, we still speak of the reproducibility of physical behaviour. Due to this discrepancy between abstract mathematical precision and unavoidable physical approximation, Max Born (1959) demands the re-formulation of the question of determinability in mechanics and, in its place, the differentiation between stable and unstable motions.

Heisenberg and Born's renunciation of the dogma of predictability may be valid for the field of atomics, but appears strange when viewed on the macroscopic scale which is directly accessible to our senses. We are, of course, aware that, in weather forecasting, we have to rely on probability estimates although the motion of the earth's atmosphere follows exactly the same physical laws as the motion of the planets. Nevertheless, the weather retains a considerable amount of randomness. The reason for this is the exceptionally high number of unknowns necessary for a detailed description of the dynamics involving correlations and feedback reactions far beyond our knowledge. Until not so very long ago, there was little reason to doubt that, in this case, at least in principle, accurate predictions would ultimately be possible. In the spirit of Laplace's demon, it was assumed that it was only necessary to gather sufficient information about the system and to process it with the necessary effort.

This mechanical conception of the world which was first shaken by quantum mechanics received a second blow as a result of an astonishing discovery, namely that even simple non-linear systems may generate irregular behaviour. We have to come to terms with the idea that such unpredictable behaviour is an intrinsic reality as it does not disappear, even after more information has been gathered. Such seemingly random behaviour which is generated in non-linear deterministic dynamics is called "deterministic chaos". The fact that deterministic laws without stochastic elements cause chaos sounds as paradoxical as the concept of deterministic chaos itself. At this point, we would like to stress that there are various manifestations of chaotic behaviour between which we must carefully distinguish. In a vessel filled with gas, the atoms fly around wildly and collide and only in this case, as Boltzmann already established, does microscopic chaos prevail. In contrast to this, macroscopic or deterministic chaos dominates when purely random oscillations occur although the laws which describe the dynamics are deterministic.

We have repeatedly mentioned the diametrically opposing conceptions of scientific knowledge: on the one hand, the idea of the atomists who stress random collision and, on the other hand, the mechanistic view of the world which is based on timeless dynamic laws. Both conceptions fail when it is a question of explaining both spatial and temporal structures, i.e. as occurring in undamped oscillations. Equilibrium thermodynamics and conventional statistical physics do not provide the methods to deal with such an oscillatory behaviour. A way out of this dilemma begins to emerge.

What is this hope based on? It is the study of the physics of nonequilibrium states and on the other hand of the theory of dynamical systems. The physics of nonequilibrium states deals with systems far from thermodynamic equilibrium where chaos prevails on the microscopic level but is completely concealed on the macroscopic level by well-organised patterns. In the second discipline, the theory of dynamical systems, it is the instabilities that play the central role. At the point of instability, the system must "choose" between various possibilities. A small, nonpredictable fluctuation decides which course the dynamical process finally takes. This occurrence of unpredictability forces us to re-think our understanding of deterministic systems and the "long-term" prognoses we have won.

Processes which are sustained by a continuous flow of energy and possibly matter from outside and which are characterised by their ordered, self-organised, collective behaviour are called evolutionary. A significant aspect of evolutionary processes is their causal coherence, although they may be interspersed with random outbursts. Anyone would reject the idea of regarding the random occurrences of a lottery as an evolutionary development. Intuitively, we demand that the current state of a system is fundamentally moulded, if not determined, by the preceding causes. This does not mean, however, that this demand for causality totally excludes random occurrences; evolutionary processes without mutation, symmetry breaking etc. are unthinkable. Random occurrences reduce the chance of exact predictions, they cannot be specified by laws, they weaken the deterministic net of cause and effect; yet it would be wrong to draw the conclusion that the pattern of reality were chaotic.

Thus, it is not only quantum physics, but also the chaos theory which throws considerable doubt on Laplace's demon of absolute determinism. "Für manchen mag dies eine Enttäuschung sein. Aber vielleicht ist ja eine Welt sogar menschlicher, in der nicht alles determiniert und nicht alles berechenbar ist, eine Welt, in der es – dank der Quantenereignisse – Zufall, und damit auch Glück, gibt, in der – weil nicht alle Probleme algorithmisch lösbar sind – Phantasie und Einfallsreichtum, Raten und Probieren, Kreativität und Originalität noch gefragt sind und in der man, wie die Chaos-Theorie zeigt, auch bei chaotischem Verhalten immer noch sinnvoll nach einfachen Grundgesetzen suchen kann." (Vollmer, 1988, p. 350)

It is certainly a result of the historical development of the sciences that the unpredictable (and thus also the chaotic state) was first explicitly suspected and then formulated mathematically in the microworld of atomic physics. This does not mean that these phenomena do not appear in our daily macroscopic life. Chaotic responses of non-linear dynamical systems are almost a matter of course to us today. We draw attention to the appearance of such phenomena in a pendulum with a large amplitude and particularly so in the case of spatial oscillations. The well-known Duffing equation is another example. A further historical example is the well-known anorganic chemical reaction of Belousov-Zhabotinsky which offers a memorable display of colour. Yet the most impressive and still the most mysterious is the onset and development of turbulence in a fluid. Research into this problem – which is highly important in technology and of decisive significance in meteorology and combustion processes – has inspired many famous physicists since Osborne Reynolds to look for a physical-mathematical solution. In spite of admirable contributions of many scientists, even Heisenberg (Heisenberg, 1948) did not succeed in solving the problem.

For fully developed turbulence, however, today's chaos theory is still not broadly enough developed. In Chapter 9, we attempt to give an overview of the current status of turbulence research and to show that further extensive physical considerations and mathematical tools are necessary to model turbulent phenomena. Apart from the difficulties involved in an adequate comprehension of a complex process, there is a further obstacle that everyone breaking new ground must surmount, even – or especially – in science. No one butWerner Heisenberg could have expressed this so memorably: "Wenn wirkliches Neuland betreten wird, kann es aber vorkommen, daß nicht nur neue Inhalte aufzunehmen sind, sondern daß sich die Struktur des Denkens ¨andern muß, wenn man das Neue verstehen will. Dazu sind offenbar viele nicht bereit oder nicht in der Lage", see (Heisenberg, 1969), p. 102.

2.2 Dynamical Systems – Examples

Before we move on to a detailed description of the chaotic behaviour of systems possessing deterministic equations of motion, we should like to lead up to this subject by beginning with four typical examples.

Our first example is a mechanical system with a single degree of freedom and no loss of energy due to friction. Let a small mass be hung on a spring and allowed to oscillate vertically (fig. 2.2.1). For small displacements, the motion of this undamped spring pendulum can be expressed by the linear differential equation in the two variables displacement x and its second derivative with respect to time t

$$\ddot{x} + w_0^2 x = 0$$

where w_0 is the frequency of the oscillation.

Equation (2.2.1) has the general solution

$$x = A \sin w_0 t + B \cos w_0 t$$

where the integration constants A and B are determined by the initial conditions.

Denoting the initial values of x and x˙ at the instant t = 0 by x0 and x˙ 0, we obtain for the displacement x

$$x = \frac{\dot{x}_0}{w_0} \sin w_0 t + x_0 \cos w_0 t$$

and for the velocity $\dot{x}$

$$\dot{x} = \dot{x}_0 \cos w_0 t - x_0 w_0 \sin w_0 t$$

Eliminating the time t from eqs. (2.2.3) and (2.2.4), we find

$$x^2 + \frac{\dot{x}^2}{w_0^2} = x_0^2 + \frac{\dot{x}_0^2}{w_0^2}$$

The family of all ellipses in the $x, \dot{x}$ -plane designated by eq. (2.2.5) is called a phase portrait. On the right of fig. 2.2.1, the displacement x as a function of time for a concrete initial condition is shown and on the left, the trajectories in the twodimensional phase space for three different initial conditions. The state variables or coordinates x and x˙ which span the phase space characterise the single-mass system uniquely.

In this idealised case where the energy is preserved, the mass returns periodically to its initial position. Also, the states of maximum deflection and zero velocity are repeated with a period of $T = 2\pi/w_0$.

In natura, it is practically impossible to realise the assumption "no loss of energy due to friction". Loss of energy, for example due to air resistance, continuously decreases the deflection of the pendulum until at some stage, the mass remains in a state of equilibrium. During the transient phase, the aforementioned elliptical trajectories in the phase space then become spirals which all end in a point, the state of equilibrium (fig. 2.2.2). This point which captures all spirals is called an attractor, in this particular case a point attractor.

When the loss of energy of the pendulum due to friction is of a viscous nature, it can be reproduced by a damping term which depends to a first approximation linearly on the velocity. The linear differential equation (2.2.1) for a conservative system then takes the following standard form, again a linear system,

$$\ddot{x} + 2\zeta w_0 \dot{x} + w_0^2 x = 0$$

where the damping factor $\zeta > 0$ controls the fading of the periodic transient response. In the case of damping $\zeta > 1$, the system tends aperiodically towards the state of equilibrium x = 0. For $0 < \zeta < 1$, the case of sub-critical damping, the amplitudes also decrease; but the motion retains qualitatively the appearance of an oscillatory process.

The loss of energy in a system can be compensated by means of the continuous supply of energy, e.g. a periodic external excitation (fig. 2.2.3).

After a certain transient phase, the trajectories in the two-dimensional phase space approach asymptotically a closed curve, the so-called limit cycle, which is run through periodically. Beside the point attractor, the limit cycle is the only possible attractor in the two-dimensional phase space.

For a harmonically excited, viscously damped oscillator, the equation of motion in linearised form becomes

$$\ddot{x} + 2\zeta w_0 \dot{x} + w_0^2 x = w_0^2 x_0 \sin w_E t$$

Here, the periodic external excitation is reproduced by a sinus term on the righthand side. Linear equations of this type are integrable and lead to a family of solutions, the individual curves of which are determined by the initial conditions. Since the dependence of a specific solution is relatively insensitive with regard to the initial conditions, small changes in the latter cause only small changes in the solution. Thus, our physical system is characterised by a "similar" connection between cause and effect.

The situation in the case of non-linear equations of motion is completely different. A simple dynamical system which leads to chaotic patterns of motion is the Duffing equation (see section 10.5). In its modified form, the restoring force is approximated by a polynomial of the third order, thus allowing, for example, the reproduction of larger deflections in an externally excited beam under shear and normal force. In this special case, the differential equation takes the form

$$\ddot{x} + 2\zeta w_0 \dot{x} - w_0^2 x + \delta w_0^2 x^3 = w_0^2 x_0 \sin w_E t$$

where the cubic term x^3 possesses a stiffening effect. Due to its simple morphology, the Duffing equation and its modifications led to numerous investigations on the behaviour of dynamical systems both in mechanics and electrical engineering (Ueda, 1980a; Ueda, 1980b; Moon and Holmes, 1979; Seydel, 1980). We stress that the non-linearity – in x, but not in time t – is a necessary condition for chaotic behaviour, though not sufficient in itself.

If the substitution $x_1 = x, x_2 = \dot{x}, x3 = t$ is carried out, the non-autonomous differential eq. (2.2.8) with the second derivative with respect to time, the righthand side of which depends explicitly on the time t, can be re-written as a system of non-linear, so-called autonomous first-order differential equations

$$
\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= -2\zeta w_0 x_2 + w_0^2 x_1 - \delta w_0^2 x_1^3 + w_0^2 x_0 \sin(w_E x_3) \\
\dot{x}_3 &= 1
\end{aligned}
$$

This autonomous set of equations is equivalent to the non-autonomous eq. (2.2.8). As a generalisation, we can state that an autonomous differential equation may be put in the form

$$\dot{\mathbf{x}} = \mathbf{F(x)}$$

where the non-linear vector function $\mathbf{F(x)}$ does not depend explicitly on the time. Thus, the introduction of additional variables transforms the non-autonomous eq. (2.2.8) with second-order derivatives with respect to time into a system of three first-order differential equations. Such a set of equations which is expressed directly in terms of displacement and velocity considerably simplifies a qualitative and quantitative discussion of the trajectories in phase space (the concept of the phase space is elucidated in section 2.3).

The phase portrait of eq. (2.2.9) for the variables $x_1, \dot{x}_1$ shows intersecting trajectories corresponding to specific choices of control parameters and initial conditions (fig. 2.2.4). Trajectories of the autonomous type of eq. (2.2.9) which do not intersect in the extended phase space with the coordinates $x_1, \dot{x}_1$ and t do so in the projection onto the $(x, \dot{x}\ )$-plane, the customary phase space. It is easy to imagine that for a non-periodic solution (i.e. an irregular or chaotic motion) for $t \to \infty$, the phase portrait will be blackened to unrecognisability in a relatively short time. For this reason, this representation must be ruled out as an indicator of chaos since it would be futile to try to identify the corresponding trajectories which remain adjacent over a long time but diverge exponentially from one another at some stage.

The most widespread form of a display of the equations of motion consists of plotting the deflection, for example, over time (fig. 2.2.5). However, to deduce a chaotic behaviour by means of this method is not realistic and is doomed to failure because the observed period of time is necessarily finite. Although the bizarre, non-periodic course of the curve in fig. 2.2.5 strongly suggests an erratic course of motion, it remains uncertain whether a periodic motion might not be established after all in case one observed a longer time interval.

It may require skill and effort to distinguish between regular and irregular behaviour on the basis of the transient response of the variables of state of non-linear dynamical models; it is, however, even more difficult to differentiate in an experimentally established transient response between background noise and deterministic chaos.

In order to assess dynamical behaviour, it is also possible to apply, apart from the representation in phase space and the temporal change of individual variables of state, the power spectrum method, well known to the engineer (see section 3.8). The recorded time series, which are evaluated in this method by means of a Fourier analysis, yield clearly defined peaks in the power spectrum diagram for periodic or quasi-periodic motions; in contrast, continuous curves or curves with a high noise level indicate chaotic or stochastic behaviour. This means that in the erratic case, the measured variables can no longer be represented as a discrete superposition of oscillations: from a critical value of this state variable in the Fourier-transformed signal onwards, a very high noise component in the spectrum emerges. Since these power spectra can easily be determined experimentally, however, they offer themselves as a means of characterising the irregular behaviour of real systems. We can illustrate this with an example taken from fluid dynamics (fig. 2.2.6). As a consequence, simple spectral analysis does not suffice as a single tool for describing erratic behaviour in chaotic physical systems. Other methods of measuring must be found in order to describe the wide spectrum of regular and irregular motions qualitatively.

In recent years, a number of procedures have been proposed for noise reduction in measured data sets (see (Kantz and Schreiber, 1997) and references cited therein). Of particular interest is the method developed recently for analysing stochastic data which is based on the theory of Markov processes and which allows the determination of the model equations underlying the experimentally measured data (Friedrich and Peinke, 1997; Friedrich et al., 2000). In contrast to other methods, this is a parameter-free approach for reconstructing the model equations, i.e. a priori, because no assumption needs to be made about the type of these equations. Moreover, in many cases different types of noise can be separated from the deterministic part (Siefert and Peinke, 2004; Böttcher et al., 2006; Lehle, 2011; Friedrich et al., 2011). As a result, one obtains stochastic differential equations which contain both a deterministic and a stochastic part. However, in the following, we restrict our considerations to dynamical systems which can be described solely by deterministic equations.

In the literature, a multitude of possible methods for characterising the occurrence of chaotic motions is presented. Apart from the power spectrum and autocorrelation, the two classic tools, we propose to concentrate on the following criteria: Lyapunov exponents, dimensions and Kolmogorov-Sinai entropy. An extensive discussion of these concepts can be found in the context of dissipative systems and attractors in Chapter 5.

The examples of mechanical systems mentioned here make it clear that the concept of the attractor plays a central role in the description of the behaviour of damped systems subject to deterministic equations of motion. We basically differentiate between two types: regular attractors and strange attractors. On the one hand, there are three classic types of motion: equilibrium, periodic motion and quasiperiodic motion. All three states are associated with regular attractors since, in the case of damping, the system tends towards one of these three states after the transient phase. The point attractor corresponds to the state of equilibrium, the limit cycle to periodic motion and the torus to quasi-periodic motion. On the other hand, there is a class of deterministic, but erratic, i.e. chaotic, motions which are not predictable if the initial conditions are subject to small fluctuations. Such long-term behaviour ($t \to \infty$) is associated with the concept of the strange attractor.

2.3 Phase Space

In the case of evolutionary processes which can be analysed quantitatively, we can record the modification of particular quantities as a function of time. These variations must stem from specific causes. If the state is defined completely at a given instant by n variables $x_1, \ldots, x_n$, the evolution of the processmay be described by a system of n ordinary differential equations

$$\begin{aligned} \frac{dx_1}{dt} &= F_1(x_1, x_2, ..., x_n) \\ \frac{dx_2}{dt} &= F_2(x_1, x_2, ..., x_n) \\ &\vdots \\ \frac{dx_n}{dt} &= F_n(x_1, x_2, ..., x_n) \end{aligned}$$

The n time-dependent variables represent physical quantities such as location, velocity, temperature, pressure etc. Not only in physics, but also in other branches of science such as biology, chemistry, economic sciences and other fields, many real processes in time can be described by systems of ordinary differential equations. If we formally introduce the column vectors

$$\mathbf{x} = \{x_1 x_2 \ldots x_n\}$$
$$\mathbf{F} = \{F_1 F_2 \ldots F_n\}$$

we can write the system eq. (2.3.1) in a simplified form (see also eq. (2.2.10))

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x})$$

where n is the number of equations defining the whole system. The fact that the system consists exclusively of first-order differential equations is not a restriction since any system of ordinary differential equations of higher order can be transformed into a system of first order by the introduction of additional variables.

We wish to stress once again that the system eq. (2.3.1) or (2.3.3) is autonomous since the right-hand side does not depend explicitly on the independent variable t.

This is not a restriction either, as each non-autonomous system of equations can be transformed into an autonomous one by introducing an additional variable xn+1 = t and the trivial relationship $\dot{x}_{n+1} = 1$.

It is very useful to represent the temporal evolution of a system in an abstract space, the phase space. It is n-dimensional and is spanned by the state variables or coordinates $x_1, x_2, \ldots, x_n$. In the phase space, the state of the system at a given time is represented by a point. This point moves with time and its velocity is specified by the vector **F**. Since the velocity **F** is known from eq. (2.3.3), the velocity field can be represented immediately in the phase space (see fig. 2.3.1). The whole set of directions specifies a direction field, reminiscent of the stream lines in a liquid.

A point chosen arbitrarily at the instant $t = t_0$ describes a curve in phase space which at each point is tangential to the vector field of velocities. The graph of the motion in the phase space is called trajectory, phase line or orbit and the total set of possible motions is denoted phase flow ϕt. The autonomous set of first-order equations (2.3.1) allows us to display the velocity field without integration directly in the phase space. We thus already obtain a first impression of the form of the solution.

Through each point of the phase space, exactly one trajectory runs. Physically speaking, this means that if a state is known at a particular instant, both the future and the past are determined by integration. This also means that trajectories which represent a unique solution can never intersect. In the special case of the mechanics of a particle where the dynamical state of a mass point is specified in three-dimensional space by its position (three space coordinates) and by its velocity (three velocity components), the phase space is six-dimensional. For a single-degree-of-freedom oscillator, the phase space degenerates to the phase plane with the components displacement and velocity (fig. 2.2.1).

2.4 First Integrals and Manifolds

In the distant past, when mathematical tradition demanded the solution of differential equations by analytical integration, the somewhat strange concept of "first integrals" was coined. What was then understood by an integral is nowadays called solution.

The definition of the first integral used here has been adopted from Arnold (Arnold, 1980). Let $\mathbf{F}$ be a vector field, and the individual components $F_1, F_2...F_n$ differentiable functions.

Definition: a function $\mathbf{I(x)}$ is called the first integral of the differential equation

$$\dot{\mathbf{x}} = \mathbf{F(x)}$$

if its so-called Lie derivative $L_{\mathbf{F}}$ (Olver, 1986) along the vector field $\mathbf{F}$ vanishes

$$L_F \equiv F_1 \frac{\partial I}{\partial x_1} + ... + F_n \frac{\partial I}{\partial x_n} = \mathbf{F}^t \frac{\partial I}{\partial \mathbf{x}^t} = \frac{\partial I}{\partial \mathbf{x}} \mathbf{F}$$

where the notation adopts the following conventions:

$$\frac{\partial I}{\partial \mathbf{x}} = \left[\frac{\partial I}{\partial x_1} \frac{\partial I}{\partial x_2} ... \frac{\partial I}{\partial x_n} \right]$$

denotes a row vector and, correspondingly,

$$\left(\frac{\partial I}{\partial \mathbf{x}} \right)^t = \frac{\partial I}{\partial \mathbf{x}^t} = \left\{ \frac{\partial I}{\partial x_1} \frac{\partial I}{\partial x_2} ... \frac{\partial I}{\partial x_n} \right\}$$

a column vector. Equation (2.4.2) implies the following characteristics of the first integral: on the one hand, the function $I(x_1, x_2...x_n)$ remains constant along each given trajectory; on the other hand, each trajectory lies on a hypersurface in the phase space. The hypersurface is defined by $I(\mathbf{x}) = C$, where C is a constant (see fig. 2.4.1 for a three-dimensional phase space). Each trajectory defined by the initial condition thus lies entirely on a smooth hypersurface or generates it.

The hypersurface defined by I($\mathbf{x}$) = C designates an (n - 1)-dimensional manifold (see also the discussion at the end of this section). If all possible values are assigned to the parameter C, this leads to a one-parameter family of manifolds which fills the phase space completely. If a first integral I is known, the equation I(x) = C for a given C can be solved with respect to xi. If this xi is substituted into the remaining (n - 1) equations of eq. (2.3.1), the whole set of equations is reduced to (n - 1) equations. The more first integrals are known, the lower the number of equations and the lower the dimension of the manifold generated by the trajectory.

In mechanics, it is the conservation theorems which often supply first integrals; unfortunately, however, no systematic rule is known which yields an easier derivation of first integrals. The Hamilton function which represents a first integral for conservative Hamilton systems is a stroke of luck. In addition, it can be shown that in the case of conservative forces, the Hamilton function – if it is not time-dependent – corresponds to the total energy, i.e. the sum of kinetic and potential energy (see also section 4.1).

It is necessary to associate many phenomena with geometrical models which cannot be described in a simple form. In the case of dynamical systems, the geometrical models form manifolds. Manifolds play an important role in the determination of the global behaviour of trajectories. For example, non-chaotic attractors which specify the asymptotic state of many dynamical systems lie on manifolds or mould them.

An n-dimensional manifold M is a topological space with the property that each point and its neighboorhood can be mapped by a one-to-one continuous bijective mapping that has a continuous inverse function, a so-called homeomorphism, onto an open subset of the n-dimensional Euclidean space E. Therefore, it is possible to use the coordinates of E as local coordinates on M, see (Abraham et al., 1993; Br¨ocker and J¨anich, 1990) and fig. 2.4.3. The underlying idea is to map globally complex, but still sufficiently smooth structures onto elementary structures in order to apply, for example, methods from differential geometry. It is well known that the earth's surface cannot be treated as a Euclidean space; measurements on it must be performed in accordance with the laws of spherical trigonometry. Yet parts of the earth can be mapped on charts of an atlas and, when approaching a chart's boundary, one has to switch to the next one.

Geometries which may not admit an overall analytical representation are considered locally in their tangential plane, i.e. the neighborhood of each point can be described by local coordinates. By joining overlapping neighborhoods, so-called charts, the entire manifold and the corresponding mapping can be described analytically. We illustrate this approach using the limit cycle in fig. 2.4.2 as a simple example of a manifold. The limit cycle of fig. 2.4.2 can be subdivided into an assembly of overlapping subsets which can be mapped onto subintervals between 0 and 2$|pi$a by a continuous one-to-one map. Thus, the ϕ-coordinate can be used directly as a local coordinate on the manifold, independent of the phase space coordinates which describe the limit cycle.

A further example of a manifold in three dimensions is the two-dimensional torus (fig. 2.4.5). For such a torus, each surface element can be mapped one-to-one onto an element in the plane. The assembly of all the overlapping surface elements once again forms a torus. In a corresponding manner, the surface elements on the associated plane can be joined together. Their assembly yields the square in fig. 2.4.6.

With scissors and glue and by rolling and bending, this square can be turned back into the torus; in this way, it becomes clear that each single point of the torus is defined by the coordinates ϕ_1 and ϕ2. A "helical line" on the torus corresponds to a general straight line on the mapping plane. Figure 2.4.6 demonstrates both coordinate lines ϕ_1 = const and ϕ_2 = const. Since both the limit cycle and the torus are hypersurfaces in the phase space and a velocity vector **F** exists at each point, they form differentiable manifolds.

2.5 Qualitative and Quantitative Approach

There are basically two contrary approaches to the study and understanding of dynamical systems. In the first case, we concretise a particular problem as a dynamical system and gather as much information as possible about its behaviour. The logical consequence is a complicated set of equations, particularly since the equations must be formulated as realistically as possible in order to incorporate all the involved effects. In the second case, we are interested in the characteristics of dynamical systems in general and not in entering into detail. Here, too, we must differentiate between two cases:

i A mathematical approximation in the classical sense which evolves and elucidates this new qualitative analysis of differential equations and develops it on the basis of assumptions and strict argumentation. Qualitative approaches are based on geometrical or topological methods which the great mathematician Henri Poincar´e (1854 – 1912) first applied to the investigation of the stability of our solar system. Topology, the study of qualitative geometry, has become an indispensible mathematical tool for describing the behaviour of dynamical systems in all their complexity.

ii An approximation or simulation in the sense of experimental mathematics, an approach made possible by the computer. In this case, the aim is to arrive at generally valid statements on dynamical systems on the basis of simple nonlinear dynamical systems, the behaviour of which is studied numerically. The choice of representative examples in dynamical systems has to be guided by intuition and experience in order to discover characteristic patterns of response and thus obtain informative results for a broader class of dynamical systems.

References

[1] John Argyris, Gunter Faust (2015) *An Exploration of Dynamical Systems and Chaos* , Springer..